A UNIVERSALLY INJECTIVE MORPHISM OF ALGEBRAIC SPACES IS RADICIAL WHEN IT IS DECENT, QUASI-SEPARATED OR LOCALLY SEPARATED

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-more-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the first line of the pinned chapter file is \input{preamble}, and the native preamble was not staged with the source, so no preamble text lives inside the chapter. The retained preamble used here is the project's single Stacks-wide preamble (the same 217-line / 8116-byte file that the already published chow and etale-cohomology cards use), copied mechanically line for line from sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Radiciality of a morphism of algebraic spaces, tested after base change to fields. Same chapter, Definition labelled definition-radicial; invoked in the proof at lines 116-119. Fixes what has to be shown for the reduced fibre (Spec(K) x_Y X)_red -> Spec(K).
- Radicial and universally injective agree for morphisms of schemes. Morphisms, Lemma labelled morphisms-lemma-universally-injective; invoked at line 132. Converts the reduced-fibre conclusion back into radiciality once the structure morphism is a morphism of schemes.
- Stability under base change and composition, and closed immersions, for the properties decent, quasi-separated, locally separated. Decent Spaces lemmas at proof lines 136-139 and Morphisms of Spaces lemmas at proof lines 157-162. Makes the uniform first paragraph applicable to each of the three properties listed in the statement.
- A nonempty decent reduced algebraic space universally injective over a field is the spectrum of a field. Decent Spaces, Lemma labelled decent-spaces-lemma-when-field; invoked at line 145. Supplies the field-spectrum conclusion that case (1) needs.

Source reference. Original lemma 03L0: sources/stacks-chow/tags.txt, source lines 4696–4696. Navigation pointer for \ref{decent-spaces-lemma-base-change-relative-conditions} used inside the counted span (line 137); the argument that consumes it is retained in the span.

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Source reference. Original lemma 03L1: sources/stacks-chow/tags.txt, source lines 4697–4697. Navigation pointer for \ref{decent-spaces-lemma-composition-relative-conditions} used inside the counted span (line 138); the argument that consumes it is retained in the span.

Source reference. Original lemma 03M5: sources/stacks-chow/tags.txt, source lines 4737–4737. Navigation pointer for \ref{decent-spaces-lemma-properties-trivial-implications} used inside the counted span (line 139); the argument that consumes it is retained in the span.

Source reference. Original lemma 047Z: sources/stacks-chow/tags.txt, source lines 5526–5526. Navigation pointer for \ref{decent-spaces-lemma-when-field} used inside the counted span (line 145); the argument that consumes it is retained in the span.

Source reference. Original lemma 0481: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 59–59. Navigation pointer for \ref{definition-radicial} used inside the counted span (line 117); the argument that consumes it is retained in the span.

Source reference. Original lemma 01S4: sources/stacks-chow/tags.txt, source lines 2325–2325. Navigation pointer for \ref{morphisms-lemma-universally-injective} used inside the counted span (line 132); the argument that consumes it is retained in the span.

Source reference. Original lemma 01IQ: sources/stacks-chow/tags.txt, source lines 1987–1987. Navigation pointer for \ref{schemes-lemma-immersion-when-closed} used inside the counted span (line 181); the argument that consumes it is retained in the span.

Source reference. Original lemma 03KL: sources/stacks-chow/tags.txt, source lines 4681–4681. Navigation pointer for \ref{spaces-morphisms-lemma-base-change-separated} used inside the counted span (line 160); the argument that consumes it is retained in the span.

Source reference. Original lemma 03KQ: sources/stacks-chow/tags.txt, source lines 4686–4686. Navigation pointer for \ref{spaces-morphisms-lemma-composition-separated} used inside the counted span (line 161); the argument that consumes it is retained in the span.

Source reference. Original lemma 042R: sources/stacks-chow/tags.txt, source lines 5338–5338. Navigation pointer for \ref{spaces-morphisms-lemma-immersions-monomorphisms} used inside the counted span (line 162); the argument that consumes it is retained in the span.

Source reference. Original lemma 03H6: sources/stacks-chow/tags.txt, source lines 4555–4555. Navigation pointer for \ref{spaces-properties-lemma-quasi-compact-affine-cover} used inside the counted span (line 170); the argument that consumes it is retained in the span.

Source reference. Original lemma 03E4: sources/stacks-chow/tags.txt, source lines 4443–4443. Navigation pointer for \ref{spaces-properties-lemma-quasi-compact-space} used inside the counted span (line 167); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a universally injective morphism of algebraic spaces over S.*

(1) *If f is decent then f is radicial.*
(2) *If f is quasi-separated then f is radicial.*

(3) *If f is locally separated then f is radicial.*

Proof. Let $\mathcal{P}$ be a property of morphisms of algebraic spaces which is stable under base change and composition and holds for closed immersions. Assume $f : X \to Y$ has $\mathcal{P}$ and is universally injective. Then, in the situation of Definition 0481 the morphism $(\mathrm{Spec}(K) \times_Y X)_{red} \to \mathrm{Spec}(K)$ is universally injective and has $\mathcal{P}$. This reduces the problem of proving

$$\mathcal{P} + \text{universally injective} \Rightarrow \text{radicial}$$

to the problem of proving that any nonempty reduced algebraic space X over field whose structure morphism $X \to \mathrm{Spec}(K)$ is universally injective and $\mathcal{P}$ is representable by the spectrum of a field. Namely, then $X \to \mathrm{Spec}(K)$ will be a morphism of schemes and we conclude by the equivalence of radicial and universally injective for morphisms of schemes, see Morphisms, Lemma 01S4.

Let us prove (1). Assume f is decent and universally injective. By Decent Spaces, Lemmas 03L0, 03L1, and 03M5 (to see that an immersion is decent) we see that the discussion in the first paragraph applies. Let X be a nonempty decent reduced algebraic space universally injective over a field K. In particular we see that $|X|$ is a singleton. By Decent Spaces, Lemma 047Z we conclude that $X \cong \mathrm{Spec}(L)$ for some extension $K \subset L$ as desired.

A quasi-separated morphism is decent, see Decent Spaces, Lemma 03M5. Hence (1) implies (2).

Let us prove (3). Recall that the separation axioms are stable under base change and composition and that closed immersions are separated, see Morphisms of Spaces, Lemmas 03KL, 03KQ, and 042R. Thus the discussion in the first paragraph of the proof applies. Let X be a reduced algebraic space universally injective and locally separated over a field K. In particular $|X|$ is a singleton hence X is quasi-compact, see Properties of Spaces, Lemma 03E4. We can find a surjective étale morphism $U \to X$ with U affine, see Properties of Spaces, Lemma 03H6. Consider the morphism of schemes

$$j : U \times_X U \longrightarrow U \times_{\mathrm{Spec}(K)} U$$

As $X \to \mathrm{Spec}(K)$ is universally injective j is surjective, and as $X \to \mathrm{Spec}(K)$ is locally separated j is an immersion. A surjective immersion is a closed immersion, see Schemes, Lemma 01IQ. Hence $R = U \times_X U$ is affine as a closed subscheme of an affine scheme. In particular R is quasi-compact. It follows that $X = U/R$ is quasi-separated, and the result follows from (2). □